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A polite, scathing, and comprehensive reply by [@jasndoc](#) and coauthors to the "ergodicity economics" of [@ole_b_peters](#).

[@NaturePhysics](#) erred in not finding a referee who would ask these basic questions of the authors, but glad a reply is out there.

[nature.com/articles/s4156...](#)

Economists' views on the ergodicity problem

Jason N. Doctor¹, Peter P. Wakker²  and Tong V. Wang² ARISING FROM O. Peters *Nature Physics* <https://doi.org/10.1038/s41567-019-0732-0> (2019)

Peters¹ Perspective on the possible role of ergodicity in the formalism of economics provides a fresh take on an established problem, yet his remedies are unjustified and give an inaccurate impression of the economics field as a whole (Supplementary Information).

Expected utility (EU) concerns choices between probability distributions over outcomes. Utility U assigns a subjective value to each possible outcome. Then the probability distribution with the highest EU value (probability-weighted average U) is chosen. This procedure involves imagining consequences that may not happen. We do such scenario planning every day, and—contrary to Peters' claims¹—we need not invoke 'parallel universes' to do so.

Peters shows that his ergodic model, when assuming ergodicity, has some mathematical properties similar to EU. Peters criticizes EU, and then all of economics, for supposedly assuming ergodicity implicitly (Supplementary Information). Yet this dynamic assumption is outside the scope of EU as a static theory. Furthermore, mathematical isomorphisms between intertemporal and other choices have been demonstrated previously².

Peters approvingly cites an experimental work by Meder et al.³, but we must note that static EU is inappropriately applied to a dynamic context there (Supplementary Information). Specifically, Meder et al.³ report inconsistencies in EU-utility measurements that, in turn, are interpreted as a falsification of EU. Such falsifications of EU are not new⁴, however, and numerous other falsifications of EU have been reported in the past⁵. Many economists have therefore used improved theories, including prospect theory⁶, which have found widespread use⁷. Peters' citation of a falsification of EU to criticize all of economics is therefore unwarranted. Also, although some ergodic predictions seem to be faithful to preference, others do not. Would a person ever prefer a process that, after three rounds, diminishes wealth from US\$10,000 to 0.5 cents over one that yields a 99.9% chance of US\$10,000,000 and otherwise US\$0? Ergodic theory predicts that this is so because the former has a higher average growth rate (Supplementary Information).

Peters suggests that intertemporal growth is the primary factor that explains economic phenomena. He suggests that the dependence of utility and other attitudinal characteristics on persons is not important, as with objects studied in physics. But kicking a person is not like kicking a rock—different people react differently. We cannot predict human choices directly from stimuli. Thousands of economic studies find interpersonal variations⁸, and they work well to explain behaviours such as undersaving and smoking, to make only two examples.

Although it is true that our consumption of economic goods develops over time, time is not the most central aspect of all our decisions. For many of our decisions, other equally ubiquitous aspects such as risks, strategy and the balancing of pros and cons are more central. Just because something is ubiquitous, it should not be confused with being explanatory; for example, we can argue

that everything consists of molecules, but it is not a reason to think that all questions in economics, geography and throughout life should be answered by molecular dynamics. We therefore caution that there is no single paradigm that is superior for answering all (economic) questions.

Economists use static EU for static decisions, when dynamics are not central. Otherwise, a dynamic model is used. Additive or multiplicative growth are specified whenever appropriate, and usually without the need to assume ergodicity. Such examples are common throughout finance and economics, including life-cycle consumption theory or Grossman's model of health⁹. In such cases, it is the consumption paths that are optimized, and not just the entire final wealth state, as in Peters' models.

We conclude with an appeal to physicists to think carefully about human behaviour to help improve economics.

Online content

Any methods, additional references, Nature Research reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41567-020-01106-x>.

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References

1. Peters, O. The ergodicity problem in economics. *Nat. Phys.* <https://doi.org/10.1038/s41567-019-0732-0> (2019).
2. Keeney, R. L. & Raiffa, H. *Decisions with Multiple Objectives* 2nd edn, Ch. 9 (Cambridge Univ. Press, 1993).
3. Meder, D. et al. Ergodicity-breaking reveals time optimal economic behavior in humans. Preprint at <https://arxiv.org/abs/1906.04652> (2019).
4. Hershey, J. C. & Schoemaker, P. J. H. Probability versus certainty equivalence methods in utility measurement: are they equivalent? *Manage. Sci.* **31**, 1213–1231 (1985).
5. Starmer, C. J. Developments in non-expected utility theory: the hunt for a descriptive theory of choice under risk. *Econ. Lit.* **38**, 332–382 (2000).
6. Kahneman, D. & Tversky, A. Prospect theory: an analysis of decision under risk. *Econometrica* **47**, 263–291 (1979).
7. Press Release: The Prize in Economic Sciences 2017 (Royal Swedish Academy of Sciences, 2017); <https://www.nobelprize.org/prizes/economic-sciences/2017/press-release>
8. L'Hartidon, O. & Vieider, F. All over the map: a worldwide comparison of risk preferences. *Quant. Econ.* **10**, 185–215 (2019).
9. Grossman, M. J. On the concept of health capital and the demand for health. *Polit. Econ.* **80**, 223–255 (1972).

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Economists' views on the ergodicity problem – Nature Physics

<https://www.nature.com/articles/s41567-020-01106-x>

This thread gives my own gloss and expansion of some points Doctor et al. raise.

Peters and co think there is a hidden assumption of economic theory: specifically, they think expected utility theory secretly assumes a mathematical property called ergodicity.

This is false.

Expected utility theory makes 4 assumptions, which are stated precisely and concisely in every graduate textbook. Ergodicity is not among them.

EU is not the kind of theory that can hide assumptions: it is like Newtonian mechanics, not like Freudian analysis.

3/

(This formulation is due to [@ShengwuLi](#).)

Indeed, basic Expected Utility Theory does not need to make any assumptions about time at all, because it is a static theory of decisions under uncertainty. EE can't deal with uncertainty at all: it wants to be all about time.

4/

For infinite income streams, they think there is ONE TRUE UTILITY FUNCTION: the long-run time-average growth rate.

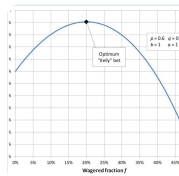
Their criterion applies only to the rare cases where that rate is deterministic: again, they can't think about uncertainty (~MuLtIpLe uNiVeRsEs~) at all.

5/

In any case, the grand theory they propose is a small special case of the (extant, rich) theory of dynamic choice, and mostly boils down to the Kelly criterion, which is a good idea from the 1950's and well-known in economics and finance.

[en.wikipedia.org/wiki/Kelly_cri...](https://en.wikipedia.org/wiki/Kelly_criterion)

6/



Kelly criterion – Wikipedia

https://en.wikipedia.org/wiki/Kelly_criterion

Kelly and contemporaries understood this nice criterion could not accommodate either the full diversity of the dynamic choices people face or of the preferences they have.

Thinking through this led to the development of modern decision theory.

7/

The EE crew, in contrast, are stuck in a grandiose exaggeration and misunderstanding of one 1950's idea, thinking that it falsifies all economics since.

Some of them, led by @oliver_b_hulme, even think they have run an experiment falsifying EUT in favor of "EE."

8/

Doctor et al. demolish this misconceived and confused experiment, beginning with the point that the authors apply static EU in a dynamic context.

9/

Doctor et al. also point out that it is trivial to falsify the predictions of ergodicity economics if you want, which is what you might expect with a theory that posits ONE TRUE UTILITY FUNCTION.

10/

More fundamentally, Doctor et al. give Peters et al. the basic economics lesson they sorely need, pointing out that the dynamics of, say, household decision-making are more diverse and complex than can be accommodated in their zero-parameter theory.

11/



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Doctor et al. have done a generous thing, though unfortunately the learning will likely be lost on the EE crew itself. They are very committed to the bit, and the idea that their magic bullet will not restart all of economics is too bitter a fact to swallow.

12/

In their commitment to the hope that they will redirect a mature field with a simple, known idea (and without engaging with current work on the same issues), they embody the main feature of scientific cranks.

13/



While it was wrong of [@NaturePhysics](#) to give a big platform to such work without soliciting an expert critique, I'm grateful that [@jasndoc](#) and coauthors have contributed their time to setting the record straight.

14/14

Many thanks to [@ShengwuLi](#) and [@PietroOrtoleva](#). Though they're not implicated in anything I say above, I am in their debt for helping me think through both what to say and how to say it!

. . .

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